



الجامعة الإسلامية العالمية شيتاغونغ
International Islamic University Chittagong
Department of Electrical and Electronic Engineering

PROJECT REPORT

Design and Implementation of AC Light Dimmer Using TRIAC and Arduino.

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COURSE TITLE : Power Electronics Sessional

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Remarks:

1. Project Title

Design and Implementation of AC Light Dimmer Using TRIAC and Arduino.

2. Abstract

This project presents the design and implementation of an Arduino Uno-based AC light dimmer circuit capable of controlling the brightness of a 220V AC incandescent lamp. The system uses phase-angle control technique to regulate the power delivered to the load. A zero-crossing detector circuit built around the PC817 optocoupler detects each AC zero crossing and sends a signal to the Arduino on digital pin 2. Upon receiving this interrupt signal, the Arduino introduces a programmable delay before firing the BTA16 TRIAC through a MOC3021 optoisolated TRIAC driver. The phase delay is controlled by a potentiometer connected to the analog pin A0 of the Arduino. The circuit ensures complete galvanic isolation between the high-voltage AC mains side and the low-voltage Arduino control side, making the design both safe and reliable. The project demonstrates the fundamental principles of TRIAC-based power control, interrupt-driven programming, and optocoupler isolation techniques widely used in industrial and domestic lighting control applications.

3.Circuit Diagram:

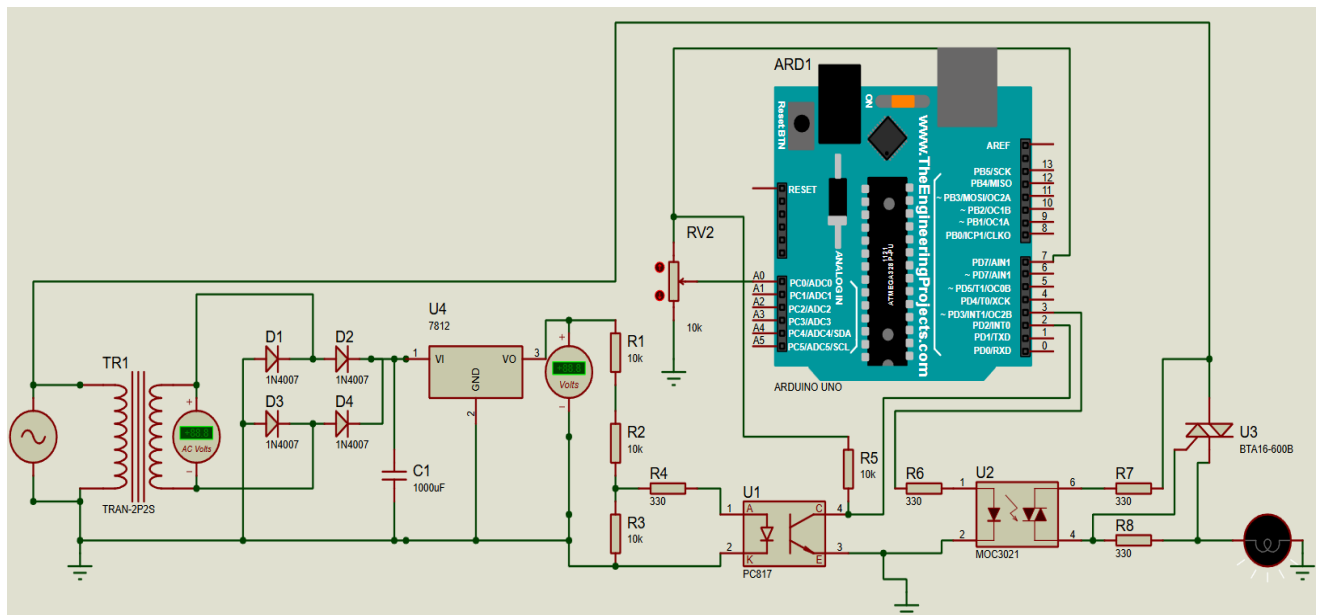


Fig. 1: Circuit Diagram of AC-Light Dimmer Using TRIAC & Arduino.

4. Component Description:

4.1 Arduino Uno: The Arduino Uno is a microcontroller board based on the ATmega328P. It operates at 5V logic and has 14 digital I/O pins, 6 analog input pins, a 16 MHz crystal oscillator, and a USB connection for programming. In this project it serves as the central controller, reading the potentiometer value, detecting zero crossings via an external interrupt, and generating the TRIAC firing pulse with precise timing.

4.2 12V Step-Down Transformer: Steps down the AC mains voltage to a lower AC voltage, which is then rectified and regulated to power the control circuitry.

4.3 1N5408 Diode: A high-current rectifier diode used to convert the stepped-down AC voltage into pulsating DC in the power supply section.

4.4 Capacitor (1000 μ F, 100 μ F): Used for filtering and smoothing the rectified DC voltage & regulator output voltage, reducing ripple to provide a stable DC supply.

4.5 LM7812: A voltage regulator IC that provides a fixed 12V DC output, used to the overall circuit.

4.6 LM7809: A voltage regulator IC that provides a fixed 9V DC output used to power Arduino.

4.7 PC817 Optocoupler: The PC817 is a 4-pin optocoupler consisting of an infrared LED and a phototransistor. It is used in the zero-crossing detector circuit to sense when the AC waveform crosses zero volts. It provides complete electrical isolation between the AC mains side and the Arduino. When the AC voltage passes through zero, the IR LED turns off and the phototransistor state changes, generating a signal on Arduino pin 2.

4.8 MOC3021 Optoisolated TRIAC Driver: The MOC3021 is a 6-pin optoisolated TRIAC driver output device. It consists of an infrared emitting diode coupled with a light-activated TRIAC. It provides isolation between the Arduino control signal and the high-voltage TRIAC gate drive circuit. When the Arduino sends a HIGH signal to pin 1 (anode) of the MOC3021, it triggers the internal TRIAC, which in turn drives the gate of the external BTA16 power TRIAC.

4.9 Potentiometer: A 10k Ω linear potentiometer is used as the brightness control input. Its wiper is connected to Arduino analog pin A0. The Arduino reads this value (0–1023) and maps it to a phase delay (0–9900 microseconds), allowing the user to smoothly control the lamp brightness from fully off to fully on.

4.10 BTA16-600B Power TRIAC: The BTA16-600B is a 16A, 600V bidirectional power TRIAC in a TO-220 package. It is capable of conducting current in both directions when triggered, making it ideal for AC power control. It has three terminals: Main Terminal 1 (MT1), Main Terminal 2 (MT2), and Gate. The TRIAC is triggered by the MOC3021 and controls the AC power delivered to the lamp load using phase-angle control.

4.11 AC Light Bulb & Holder: The AC-powered load being controlled, mounted in a holder for safe connection to the switching circuit.

5. Circuit Operation:

The circuit operates in multiple stages.

The power supply stage: The 220V AC mains supply is stepped down to 12V AC by the transformer TR1. The four 1N5408 diodes (D1–D4) in a bridge rectifier configuration convert this to pulsating DC. The 1000 μ F capacitor (C1) smooths the DC, and the 7812 voltage regulator provides a stable 12V output to power the Arduino.

Zero Crossing Detection: A portion of the rectified AC is fed through a voltage divider (series 10k Ω resistors) to limit the current into the PC817 IR LED through a 330 Ω resistor. Each time the AC waveform crosses zero volts, the current through the IR LED drops to zero, causing the phototransistor to switch states. This transition generates a pulse on Arduino digital pin 2, which is configured as an external interrupt (INT0). The Arduino detects this RISING edge and immediately calls the interrupt service routine (ISR).

Phase Angle Control: Inside the ISR, the Arduino waits for a delay period (stored in the variable "brightness") before firing the TRIAC. This delay represents the phase angle. A longer delay means the TRIAC fires later in the half-cycle, delivering less power to the load (dimmer). A shorter delay means earlier firing and more power (brighter). The delay is derived from the potentiometer reading, mapped between 0 and 9900 microseconds (corresponding to nearly a full half-cycle of 50Hz AC which is 10,000 μ s).

TRIAC Firing: After the phase delay, the Arduino sends a HIGH pulse on digital pin 3 to the MOC3021 anode (via 330 Ω resistor R1). This activates the MOC3021 internal TRIAC, which drives the gate of the BTA16 power TRIAC. The BTA16 turns ON and conducts the AC current through the lamp until the next zero crossing, at which point it naturally turns OFF (TRIAC self-commutation). This process repeats every half-cycle, creating a phase-controlled AC waveform that controls lamp brightness.

Isolation: Both the PC817 (zero crossing side) and MOC3021 (TRIAC drive side) provide complete optical isolation between the high-voltage 220V AC mains and the low-voltage 5V Arduino control circuitry. This ensures that any fault on the AC side has no direct electrical path to the microcontroller.

6. Cost Calculation:

Sl.No	Component	Quantity	Approx. Unit Price	Total
1	Arduino Uno R3	1	1000	1000
2	12V Step-Down Transformer	1	220	220
3	LM7812 Voltage Regulator	1	20	20
4	LM7809 Voltage Regulator	1	20	20
5	Potentiometer (10k)	1	15	15
6	1N5408 Diode	4	5	20

Sl.No	Component	Quantity	Approx. Unit Price	Total
7	Capacitor	1	10	10
8	PC817	1	10	10
9	MOC3021	1	60	60
10	BTA16	1	40	40
11	AC Light Bulb	1	40	40
12	Light Bulb Holder	1	40	40
13	Resistors	8	1	8
14	Copper Clad Board	1	80	80
Total				1583

7. Project Implementation Video Link:

https://drive.google.com/file/d/1IUKgNE4me0Y49SibncLaMK0_OQCN_9DB/view?usp=sharing

8. Project Picture:

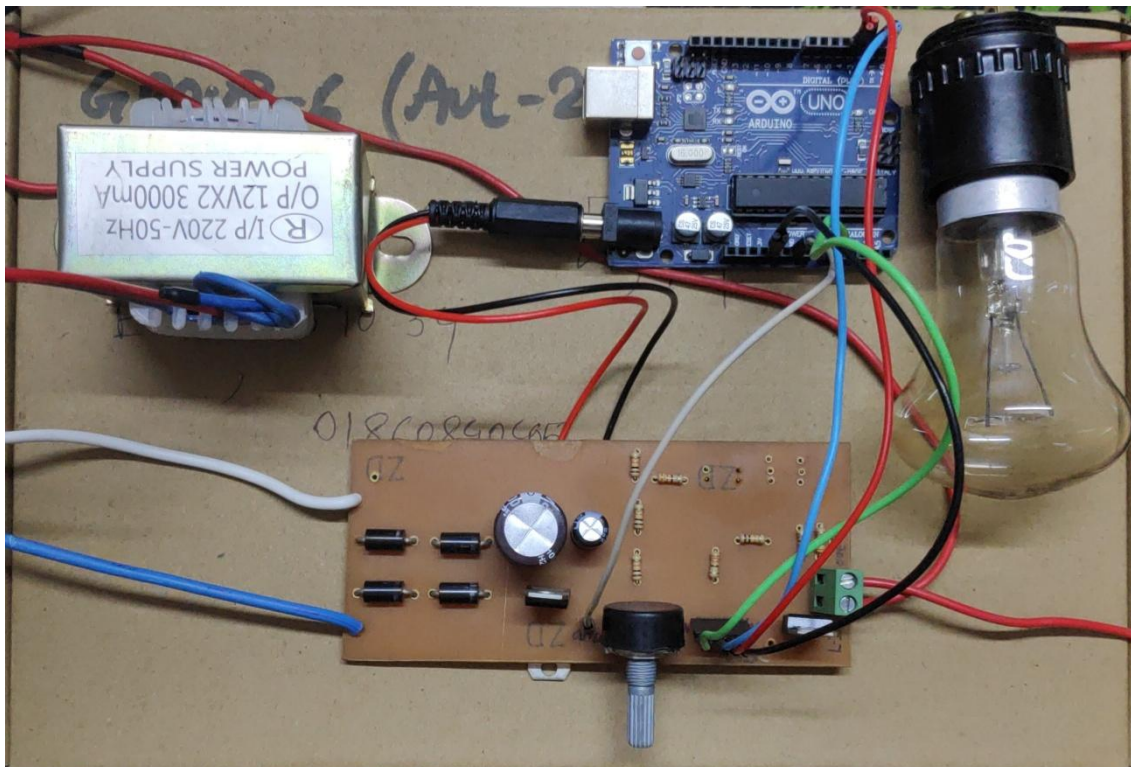


Fig. 2: Hardware implementation of AC- Light Dimmer Using TRIAC & Arduino.